CANDIDATE NAME: ______________________________________ INDEX NUMBER _________

CG _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIM EXAMINATION 2016

BIOLOGY PAPER 1
Higher 2

Thursday
22 September 2016

1 hour 15 minutes

Additional materials:
OTAS Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in this paper. Record your choice in **2B pencil** on the OTAS sheet provided.

At the end of examination, submit the question paper and MCQ OTAS sheet separately.

This question paper consists of 28 printed pages excluding this cover page.

Section A [40 marks]

Answer all questions on the OTAS provided.

1. Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?

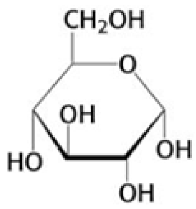
   A Presence or absence of 180^{o} rotation of alternate subunits.

   B Presence or absence of branching in the macromolecule

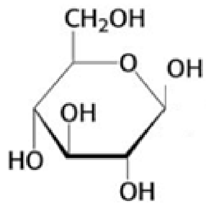
   C Subunits are either monosaccharides or amino acids.

   D Different tendency of macromolecule to form hydrogen bonds with water.

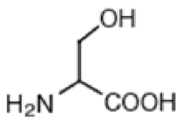
2. Which of the following combination of polymer, monomer and bond formed between monomers is correct?



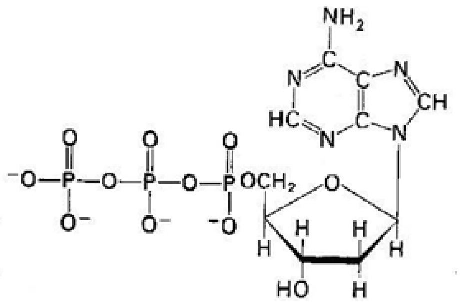
U



X



Y



Z

	starch	*cellulose*	*polypeptide*	*polynucleotide*
A	**X**, β -1,4 glycosidic bond	**U**, α -1,4 glycosidic bond	**Z**, ester linkage	**Y**, disulphide linkage
B	**U**, α -1,4 glycosidic bond	**X**, β -1,4 glycosidic bond	**Y**, peptide bond	**Z**, phosphodiester linkage
C	**Z**, peptide bond	**X**, hydrogen bond	**Z**, ionic bond	**U**, hydrogen bond
D	**X**, ionic bonds	**Y**, peptide bond	**U**, hydrogen bond	**Z**, α -1,6 glycosidic bond

3. Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it was applied to the roots of the plant. As the first step on finding out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated. Which of the following statements are consistent with the scientist's observations?

I. The toxin was stored in the central vacuole.
II. The toxin cannot cross the membrane of the organelle in which it is stored.
III. The toxin was stored in chloroplast.
IV. The toxin is likely to be lipid-soluble.
V. The toxin may be an enzyme.

A **I, II** and **V**

B **I, IV** and **V**

C **II, III** and **IV**

D **III, IV** and **V**

4. Which of the following is/are the most likely consequence/(s) for a cell lacking functional lysosomes?

(i) The cell becomes crowded with undegraded wastes.
(ii) The cell dies because its ATP-synthesizing mechanisms are missing.
(iii) The cell dies from a lack of enzymes to catalyze metabolic reactions.
(iv) The cell is unable to reproduce itself.
(v) The cell is unable to grow to a mature size and always remains small.

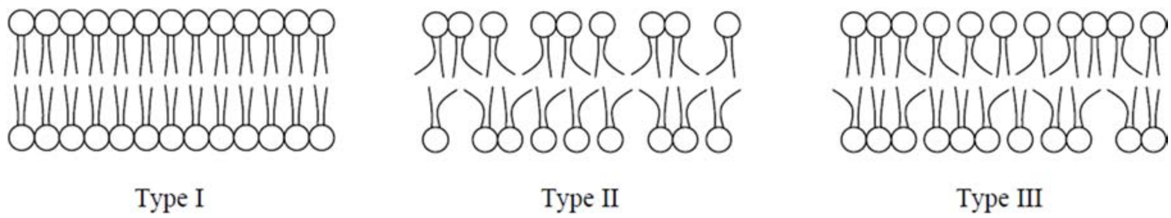
A **(i)** only

B **(i)** and **(v)**

C **(ii)** and **(iv)**

D **(iii)** and **(iv)**

5. A mutated strain of the bacterium *Escherichia coli* was found to be incapable of incorporating unsaturated phospholipids into its plasma membrane. Which of the following correctly depicts and describes the membrane of such a bacteria?



A The membrane would appear as Type I and would be more fluid at low temperatures.

B The membrane would appear as Type I and would be less fluid at low temperatures.

C The membrane would appear as Type II and would be more fluid at low temperatures.

D The membrane would appear as Type III and would be no different from normal bacterial membranes.

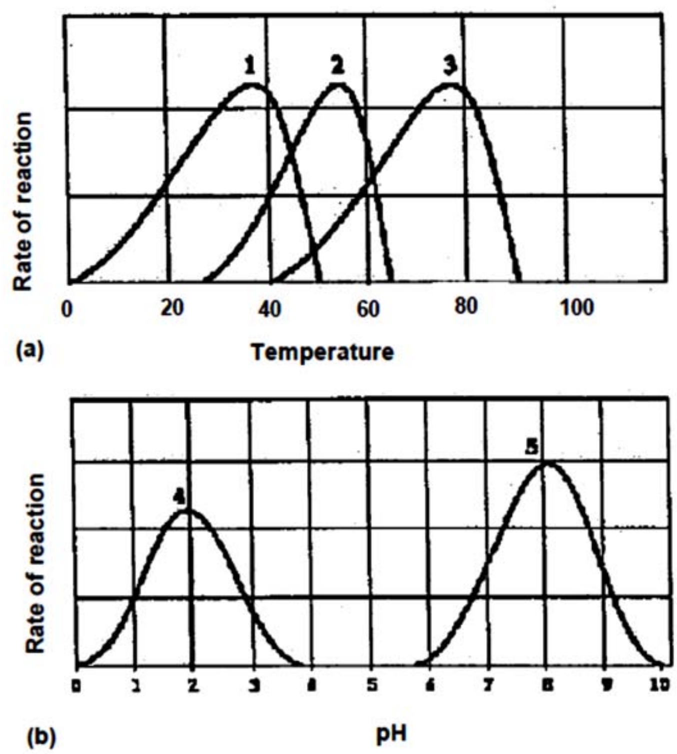
6. When mucus is secreted from a goblet cell in the trachea, these events take place.

(i) addition of carbohydrate to protein
(ii) fusion of the vesicle with the plasma membrane
(iii) secretion of a glycoprotein
(vi) separation of a vesicle from the Golgi apparatus

What is the sequence in which these events take place?

A (i), (vi), (ii), (iii)
B (i), (vi), (iii), (ii)
C (vi), (i), (ii), (iii)
D (vi), (i), (iii), (ii)

7. The following graphs show the activities of different enzymes (1-5) under different conditions:



Which statement is a correct explanation for the rate of reaction of different enzymes?

A Kinetic energy of enzyme 3 and its substrate is fastest at 75°C.

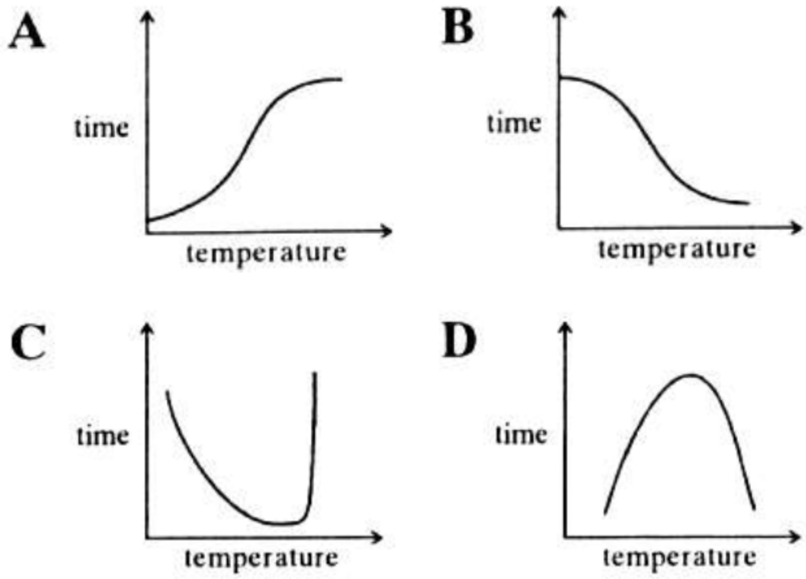
B At pH 2, most of the R groups at the active site of enzyme 4 are all negatively charged.

C At pH 8.1, substrate is bonded to the active site of enzyme 5 by hydrogen bonds only.

D At 60°C, several hydrogen bonds between R groups of enzyme 2 are broken.

8. In an investigation to determine the effect of temperature on the activity of an enzyme, the time taken for all the substrates to disappear from a standard solution was recorded.

Which graph shows the result of this investigation?



9. Which of the following statements about meiosis is **false**?

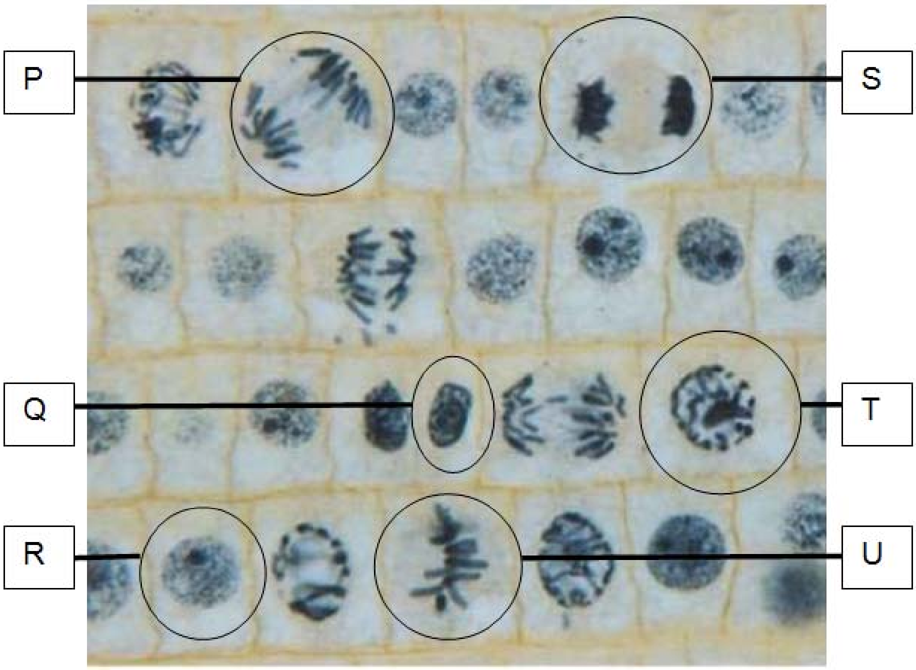
A Sister chromatids are separated at anaphase II

B Homologous chromosomes are separated in anaphase I.

C Cells at the beginning of Meiosis II are haploid

D Homologous chromosomes are paired on the metaphase plate in metaphase I.

10. The photographs below show a section of the onion root tip. The cells are at different stages of mitosis.



Which of the following shows the correct sequence of events that occurs in these cells?

A R, T, U, P, S, Q

B T, U, P, S, Q, R

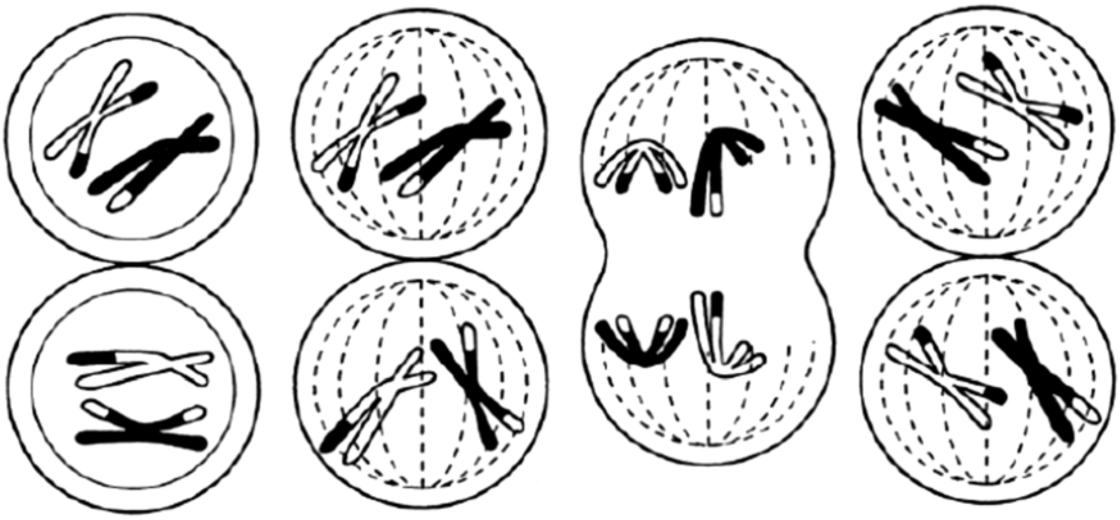
C Q, T, P, S, U, R

D T, R, P, S, U, Q

11. The diagram shows anaphase I of meiosis.

Which diagram shows metaphase II as meiosis continues in this cell?

A B C D



12. For a double-stranded DNA, which of the following base ratios always equals 1?

I. (A + T) / (G + C)
II. (A + G) / (C + T)
III. C / G
IV. (G + T) / (A + C)
V. A / G

A **I** and **III**

B **I**, **II** and **III**

C **II**, **III** and **IV**

D **III, IV** and **V**

13. The diagram below is a template strand of a DNA molecule.

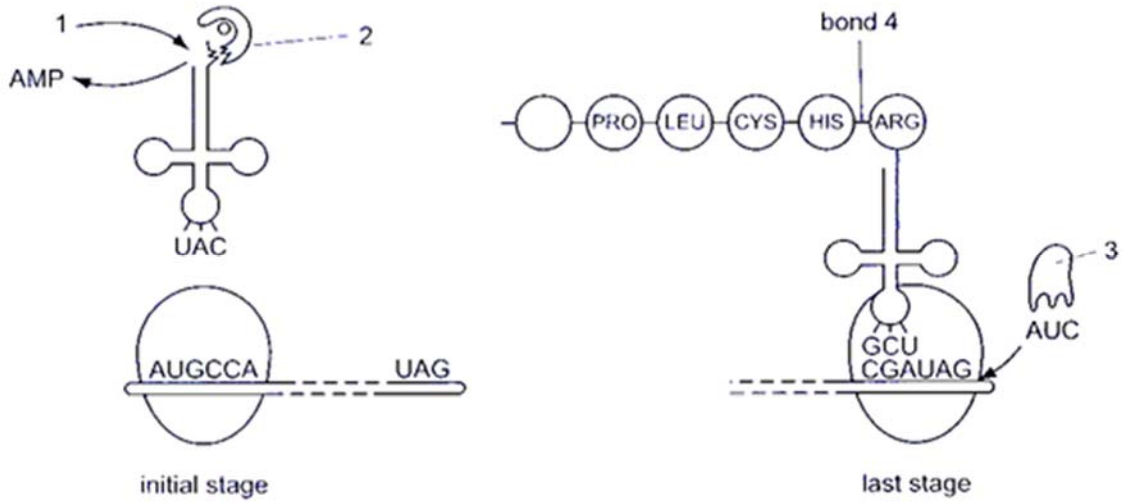
Promoter (18 nucleotides in length)	TAC	DNA sequence of 330 nucleotides	ATC

The number of amino acids in the protein coded by this template strand is

A 110 **B** 111 **C** 112 **D** 118

14. A number of molecules other than tRNA and mRNA are involved during translation.

The diagram shows some of these molecules and nucleotides in the codon and anticodon positions.



Which of the following is correct?

	1	2	3	4
A	ADP	Aminoacyl tRNA synthetase	Amino acid	Hydrogen bond
B	ADP	Amino acid	Translation releasing factor	Hydrogen bond
C	ATP	Aminoacyl tRNA synthetase	Aminoacyl tRNA	Peptide bond
D	ATP	Aminoacyl tRNA synthetase	Translation releasing factor	Peptide bond

15. With reference to a single eukaryotic gene, which of the following molecules contains the fewest number of nucleotides?

A A single strand of the original DNA segment

B A primary RNA transcript made from the original DNA segment

C A single strand of the original DNA segment after a point mutation

D A single strand of the complementary DNA (cDNA) made from the mature mRNA

16. DNA methylation is known to silence genes because it prevents transcription factors from binding. Which of the following best explains this phenomenon?

A DNA methylation modifies the shape of the transcription factor.

B DNA methylation prevents dimerization of DNA binding proteins.

C DNA methylation modifies the shape of the DNA element where the transcription factor binds.

D DNA methylation causes acetylation of histone proteins which causes heterochromatin to be formed.

17. Which of the following statements are true of the end-replication problem of linear DNA?

1. When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 3' end.
2. Telomerase prevents the end-replication problem from occurring.
3. Repeated rounds of replication produce shorter and shorter DNA molecules.
4. Prokaryotes do not have the end-replication problem.

A 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

18. Rhabdoviruses infect human cells. The genome of rhabdoviruses consists of a single-stranded RNA molecule whose sequence is complementary to the RNA sequence which functions as a messenger RNA. How is the "+" messenger RNA produced in such cells by rhabdovirus?

A Host cell RNA polymerase activity
B One portion of the infecting RNA is directly translated by host cell ribosomes.
C Reverse transcriptase activity
D The infecting virus particle contains an RNA-dependent RNA polymerase.

19. Some events that take place during generalized transduction are listed below.

I Bacterial host DNA is fragmented
II Bacterial DNA instead of viral DNA may be packaged in a phage capsid
III Recombination between donor DNA and recipient DNA
IV Phage infects a bacterial cell
V Phage DNA and proteins are made
VI Release of progeny virus

Which sequence of events is most accurate in describing generalized transduction?

A **IV, I, III, V,VI, II**

B **IV, I, V, II, VI,III**

C **IV, III, I, V, II,VI**

D **IV, V, I, III, II,VI**

20. A mutation that makes the regulatory gene of an inducible operon non-functional would result in

A continuous transcription of the operon’s genes.
B reduced transcription of the operon’s gene.
C accumulation of large quantities of a substrate for the catabolic pathway controlled by the operon.
D irreversible binding of the repressor to the promoter.

21. The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are made due to the action of the enzyme tyrosinase in cells called melanocytes. A normal level of activity of the enzyme leads to the production of red pigment, whilst a high activity allows only black pigment production. The activity of the enzyme is increased by melanocyte stimulating hormone (MSH), which binds to an MSH receptor.

The receptor is coded for by the **E** locus, which has the three alleles, $\mathbf{E^D}$, $\mathbf{E^A}$ and **e**. $\mathbf{E^D}$ is insensitive to protein A which blocks MSH receptor. $\mathbf{E^A}$ is sensitive to protein A which blocks MSH receptor. No receptor is produced by the recessive allele, **e**.

The dominant allele of a second gene, the **A** locus, codes for a protein which binds to and blocks the MSH receptor coded for by $\mathbf{E^A}$, thus preventing stimulation of tyrosinase activity in a melanocyte.

What would be the coat colours of cattle with the following genotypes?

	eeAa	$\mathbf{E^Aeaa}$	$\mathbf{E^DeAa}$	$\mathbf{E^AE^AAa}$
A	Red	Black	Black	Red
B	Red	Black	Red	Red
C	Black	Red	Red	Black
D	Red	Red	Black	Black

22. Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria. What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria?

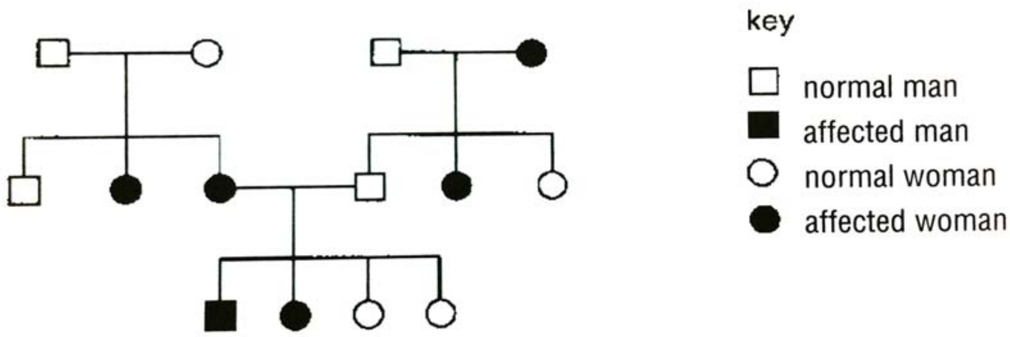
A 1 in 16

B 1 in 8

C 3 in 16

D 3 in 8

23. The family tree shows the inheritance of a skin condition.



What is the genetic basis of the skin condition?

A autosomal dominant

B sex-linked dominant

C autosomal recessive

D sex-linked recessive

24. In the board bean, a pure-breeding variety with green seeds and black hilums (the point of attached of the seed to the pod) was crossed with a pure-breeding variety with yellow seeds and white hilums. All the F_1 plants had yellow seeds and white hilums. When these were allowed to self-fertilise, the plants of the F_2 generation produced the following seeds.

Yellow seeds and white hilums	89
Yellow seeds and black hilums	28
Green seeds and white hilums	25
Green seeds and black hilums	45

A chi-squared (χ^2) test was performed to test the significance of the difference between the observed and expected results.

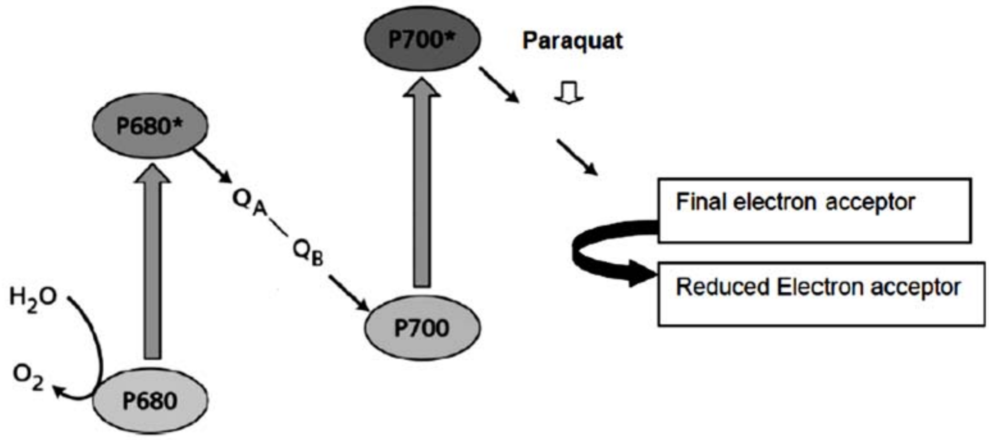
$$\chi^2 = \sum \frac{(O-E)^2}{E} \qquad \nu = c - 1$$

degrees of freedom	*p = 0.5*	*p = 0.1*	*p = 0.05*	*p = 0.01*	*p = 0.001*
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46

Which combination correctly describes the results of the χ^2 test?

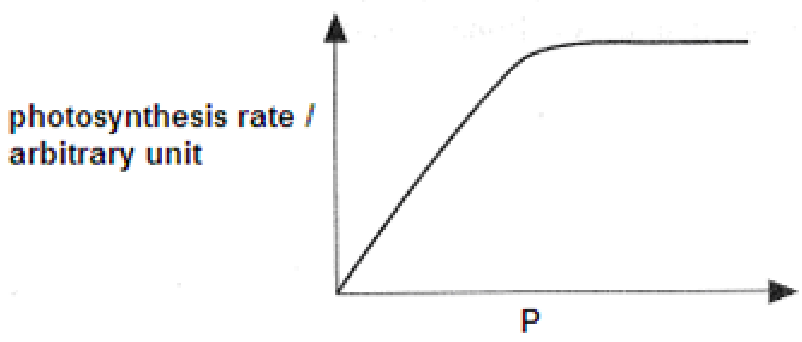
	number of degrees of freedom	*probability*	*these are two pairs of segregating alleles at two loci*
A	3	< 0.05	yes
B	3	< 0.05	no
C	4	> 0.05	yes
D	4	< 0.05	no

25. Paraquat is a poison that disrupts electron transport at the position indicated in the diagram. Paraquat was added to isolated chloroplasts. Which of the following correctly represents the outcomes in the presence of paraquat?



A Both light dependent reactions and Calvin cycle will not be able to proceed.
B Only Calvin cycle can proceed.
C Only light dependent reactions can occur.
D Both light dependent reactions and Calvin cycle can still proceed as per normal.

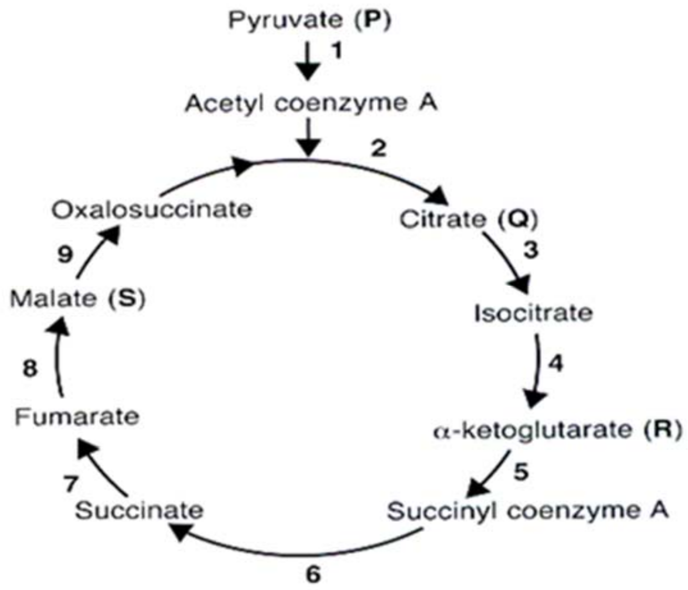
26. The following graph shows the relationship between the rates of photosynthesis with environmental factor **P**.



P most probably represents
1. carbon dioxide concentration.
2. oxygen concentration.
3. light intensity.
4. temperature.

A ***1*** and ***2*** **B** ***1*** and ***3*** **C** ***1*** and ***4*** **D** ***1***, ***3*** and ***4***

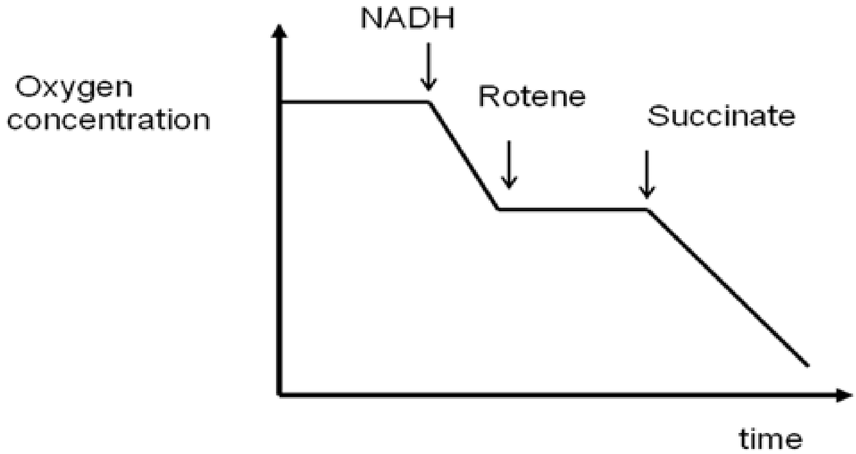
27. The diagram below shows the Krebs Cycle.



At which stages are carbon atoms removed?

A 1, 4 and 7
B 1, 4 and 5
C 2, 4 and 5
D 4, 5 and 8

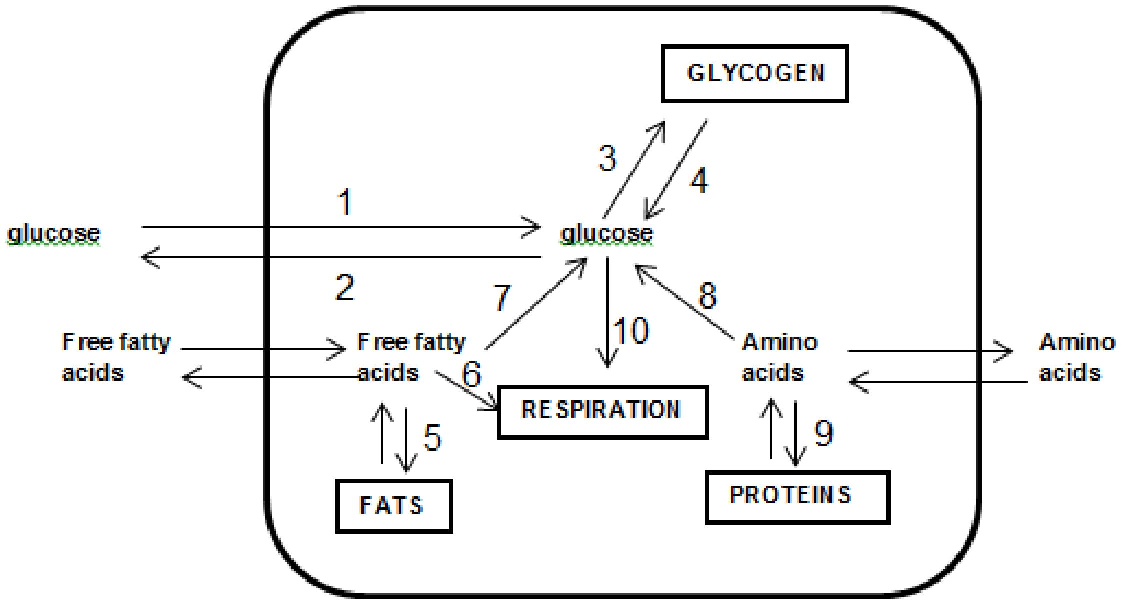
28. A suspension of mitochondria was prepared in a buffer containing ADP and inorganic phosphate (Pi). The oxygen concentration in the buffer was monitored carefully and recorded as shown below. At the times indicated, a specific reagent was added to the buffer. Throughout the experiment, the concentrations of ADP and Pi were in excess.



Which one of the following shows correctly from the highest to the lowest, the rate of ATP production after the addition of the three chemicals?

	Highest rate of ATP production	→	***Lowest rate of ATP production***
A	NADH	Succinate	Rotene
B	Succinate	NADH	Rotene
C	Rotene	NADH	Succinate
D	Rotene	Succinate	NADH

29. The diagram below shows some biochemical pathways in a liver cell. During which processes will the hormone glucagon exert an effect?



A (1), (3), (5), (9)

B (1), (4), (8), (10)

C (2), (7), (9), (10)

D (4), (6), (7), (8)

30. Rhodopsin, a light sensitive pigment that is present on the rods in the eyes. Rhodopsin is a G protein coupled receptor. The flow chart below shows how the chain of reactions that occur when rhodopsin absorbs a photon of light.

One photon of light ⇨ 1 rhodopsin

⇩ (1)

500 G protein molecules are activated

⇩ (2)

500 cyclic GMP phosphodiesterase molecules are activated

⇩ (3)

10^5 molecules cyclic GMP being hydrolysed

⇩ (4)

250 cation channels close

⇩ (5)

Membrane potential is altered by 1mV

⇩ (6)

Signal relayed to the brain

Identify the stages where *amplification* and *cellular responses* have occurred.

	Amplification	***Cellular response***
A	1, 2, 3, 4, 5	6
B	1, 3	1, 2, 3, 4, 5, 6
C	1, 3	5
D	2, 3	5

31. A neurone is undergoing relative refractory period. Which of the following statements is true?

A Membrane potential is less negative than resting potential.

B There is delayed closing of voltage-gated potassium channels.

C It is possible to initiate another action potential if a weaker stimulus is applied.

D It is impossible to initiate another action potential regardless of the stimulus strength.

32. Tetrodotoxin, a puffer fish toxin, blocks voltage-gated sodium channels. Black widow spider's venom causes the voltage-gated calcium channels to be constantly open. Crotoxin binds irreversibly to acetylcholine receptors. What will happen to the transmission of nerve impulses if each toxin is applied?

	Tetrodotoxin	**Black widow spider's venom**	**Crotoxin**
A	block action potentials along axon	reduce transmission of impulse across synapse	increase transmission of impulse across synapse
B	increase transmission of impulse across synapse	reduce transmission of impulse across synapse	block action potentials along axon
C	block action potentials along axon	increase transmission of impulse across synapse	reduce transmission of impulse across synapse
D	reduce transmission of impulse across synapse	block action potentials along axon	increase transmission of impulse across synapse

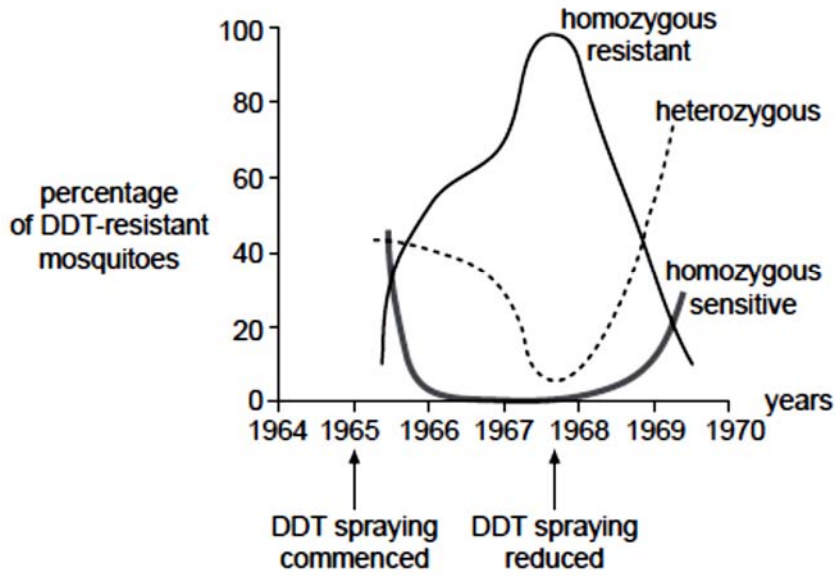
33. Members of two different species possess a similar-looking structure that they use in a similar fashion to perform the same function. Which information would best help distinguish between an explanation based on homology versus one based on convergent evolution?

A The two species live at great distance from each other.
B The two species share many proteins in common, and the nucleotide sequences that code for these proteins are almost identical.
C The sizes of the structures in adult members of both species are similar in size.
D Both species are well adapted to their particular environments.

34. In the mid-1960s, DDT was widely used as an insecticide against mosquitoes. The sensitivity to insecticide in mosquitoes is determined by a single gene that has two alleles.

allele **1** : resistant to DDT
allele **2** : sensitive to DDT

Over several years genotypic frequencies were measured in a population of mosquito larvae. The graph below shows the results.



Analysis of the graph reveals that in the population

A when spraying levels declined, heterozygous advantage occurred.

B there were no alleles for sensitivity present in the population in 1967.

C the number of alleles for resistance was equal to the number for sensitivity in 1966.

D the homozygous resistant genotype was unable to produce offspring at low spraying levels.

35. Which of the following statements is/are true of genetic drift?

(i) Genetic drift requires the presence of variation.

(ii) Genetic drift can result in the loss of beneficial alleles.

(iii) Founder effect and genetic bottleneck drive genetic drift

(iv) Genetic drift occurs in small population only

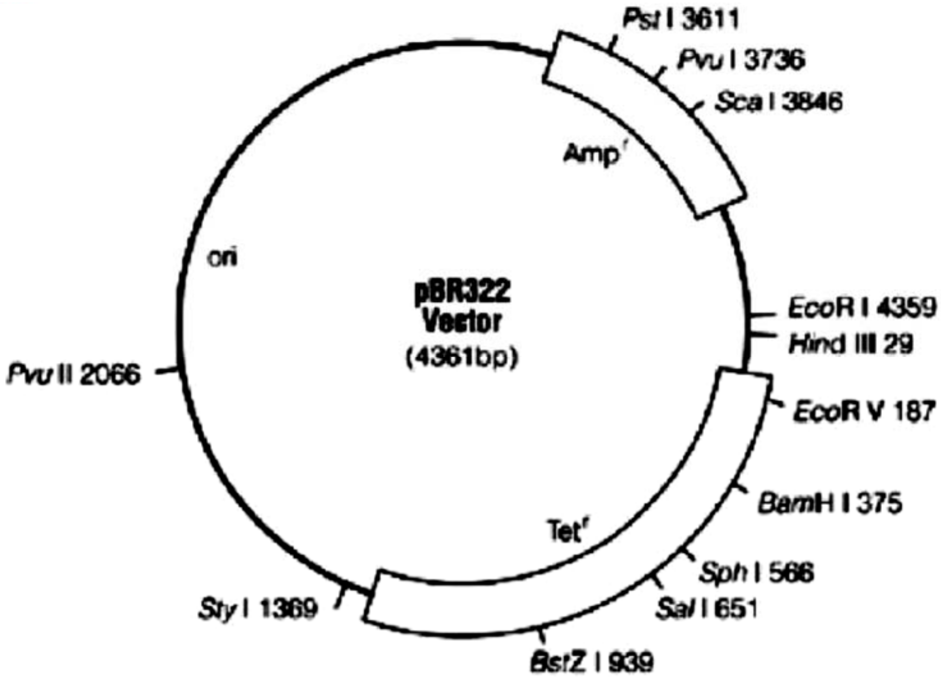
A (i) and (iv) only

B (ii) and (iii) only

C (i), (ii) and (iii) only

D (ii), (iii) and (iv) only

36. pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction endonuclease BamHI.



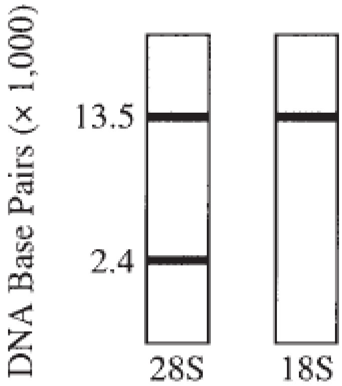
Following transformation, bacterial cells were grown in four different media, as shown below:

I nutrient broth plus ampicillin
II nutrient broth plus tetracycline
III nutrient broth plus ampicillin and tetracycline
IV nutrient broth without antibiotics

In which media would bacterial cells that contain the recombinant plasmids grow in?

A I and II
B I and III
C I and IV
D IV only

37. The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA. In the autoradiogram on the left, the probe was 28S rRNA; at the right, the probe was 18S rRNA.

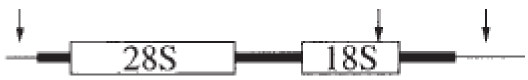


If the arrows in the following map show the location of the restriction sites of this restriction enzyme, which map *best* explains the results shown above?

A

B

C



D

38. Which of the following statements are **true** about all stem cells?

I Stem cells can be induced to differentiate by environmental signals.
II Stem cells are easily isolated and propagated
III Stem cells are able to develop into whole organisms if implanted into the womb
IV Stem cells make more stem cells under appropriate conditions.

A I and IV

B II and III

C I, III and IV

D All of the above

39. Non -viral *ex vivo* transfer of a gene encoding coagulation factor VII was performed using fibroblast cells isolated from patients suffering from severe haemophilia A.

What is the sequence of events for the *ex vivo* transfer of the gene encoding coagulation factor VIII?

1 Transfection with plasmids containing the gene encoding coagulation factor VII

2 Implantation of cells into patients

3 Isolation of fibroblasts

4 Selection and cloning of cells expressing coagulation factor VII

A 2, 3, 1, 4

B 3, 1, 4, 2

C 3, 2, 4, 1

D 3, 4, 2, 1

40. Maize varieties are being developed in which the leaves produce proteins that are toxic to insects. The DNA coding for these toxic proteins was inserted into a maize chromosome via a bacterial plasmid. Many people are opposed to this process.

Which objection is **not** biologically valid?

A Beneficial insects may be killed if they eat genetically modified maize.

B Genes for antibiotic resistance are present in plasmids and these genes may be passed to harmful bacteria.

C Hybridisation may transfer the bacterial genes from maize to weeds, giving the weed species new and harmful characteristics.

D Mutations may be caused in cattle or humans that eat the genetically modified maize.

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